Image Tampering Detection Using Multi-Feature Scoring and PCA-Based Classification

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***Abstract*—Image tampering and alteration has become an increasingly sophisticated threat due to the ease of use of digital manipulation tools. This research endeavor presents a computationally simple methodology for identifying tampered images using a multi-feature scoring system implemented in MATLAB. Six analytical techniques are used, namely, metadata inspection, error level analysis (ELA), JPEG quantization analysis, noise inconsistency analysis, DFT/DCT frequency energy analysis, and lighting inconsistency detection are utilized to assess the image integrity. Principal component analysis (PCA) is employed to consolidate the aforementioned feature vectors into a unified risk score for classification. This experimental setup tested multiple images, ranging from originals to altered ones. The results demonstrate that our approach provides an efficient classification of images into low, medium, and high risk categories.**

***Index Terms*—Fake Image Detection, Image Forensics, MATLAB, PCA, Error Level Analysis, JPEG Artifacts, Metadata Analysis, Frequency Domain, Lighting Analysis, PCA, Deep Learning.**

I. Introduction

A. Problem Statement

In today's information-driven society, images serve as powerful carriers of communication. With the rise of accessible editing software such as Adobe Photoshop, GIMP, and other mobile-based applications, image manipulation has become increasingly sophisticated,while being easy to implement and difficult to detect with the naked eye. This raises significant concerns about trust, integrity, and security in digital communication.

B. Motivation

Our motivations include misinformation campaigns, digitally manipulated celebrity images, falsified crime scene evidence, and AI-generated deepfakes which have contributed to a growing need for forensic tools that can detect altered images. Although numerous techniques have been developed to detect specific types of manipulations, there is limited research on how to combine diverse techniques into a unified pipeline. This paper seeks to fill that gap by offering a MATLAB-based solution that is both modular and scalable.

C. Scope of Work

Our research focuses on identifying image which have been tampered with through integration of six forensic techniques. The study emphasizes efficiency, interpretability, and applicability to a wide range of real-world scenarios, such as journalism, legal investigations, and cybersecurity.

D. Relevance

Despite the rise of deep learning approaches and complex machine learning algorithms in the past decades, handcrafted feature-based methods remain relevant due to their interpretability, lower computational requirements, and effectiveness in low, missing and inaccurate data or constrained environments. Combining these features with statistical and machine learning models, such as PCA, offers a robust and explainable framework. Recent trends emphasize fusion approaches that integrate multiple forensic cues, such as noise, lighting, artifacts in the frequency domain and compression inconsistencies, to achieve more comprehensive tamper detection.

E. Contributions

The primary contributions of this research are as follows:

- Development of a MATLAB pipeline that incorporates six image forensic techniques.
- Design of a unified scoring mechanism that simplifies complex outputs into risk levels.
- Application of PCA for dimensionality reduction and effective classification.
- Experimental validation across ten diverse image samples.

F. Organization of the Paper

Subsequent sections of this paper are organized as follows: Section 2 reviews relevant literature and related work. Section 3 describes the system overview and its stages. Section 4 describes the methodology, including feature extraction of each of the 6 dimensions, PCA application, and classification. Section 5 presents and discusses the results. Section 6 talks about the future work which can be done in this realm of image processing application. Section 7 concludes the study and suggests future improvements.

II. Literature Review

The domain of image forgery detection has evolved over the past decade. A few notable contributions are discussed below:

- Epstein et al. (2023) developed an online detection framework for AI-generated images that adapts over time, addressing challenges related to generalization across unseen image generators.
- Wang et al. (2023) proposed a Gated Expert Convolutional Neural Network (GE-CNN) to distinguish AI-generated images from real ones. Their architecture uses multiple specialized subnetworks and a gating mechanism for selective feature learning.
- Yao et al. (2023) introduced the Inconsistency-Aware Wavelet Dual-Branch Network (IWBNet), a method that utilizes both spatial and frequency features to effectively detect manipulated facial images.
- Cui et al. (2023) presented an End-to-End Reconstruction-Classification Network (E2E-RCN) that combines reconstruction and classification branches to better retain forensic traces in manipulated images.
- Zhao et al. (2022) introduced a method based on statistical analysis of mantissa value distributions in digital images, revealing patterns unique to authentic versus manipulated content.
- Singh et al. (2021) designed a handcrafted edge-based approach that combines textural, frequency, and statistical characteristics to identify forged regions in images.
- Qureshi et al. (2020) proposed a two-stage CNN-Keypoint-based hybrid model for copy-move forgery detection. The approach combines keypoint localization with deep feature learning to improve robustness against geometric transformations.
- Bappy et al. (2019) proposed a hybrid architecture combining CNN and LSTM networks to capture both local and contextual manipulation cues in high-quality forged images.
- Zhou et al. (2018) introduced a multi-task CNN to perform both localization and classification of manipulated regions, improving the precision of detection.
- Cozzolino et al. (2017) developed an autoencoder-based solution to detect image forgeries by modeling pristine patch distributions and identifying anomalies.
- Mahdian and Saic (2010) reviewed passive methods, categorizing them into copy-move, splicing, and resampling detection.
- Farid (2009) emphasized lighting inconsistencies and geometric cues in manipulated images.
- Fridrich et al. (2003) introduced the concept of analyzing JPEG compression artifacts to detect tampering.

III. System Overview

The proposed image forgery detection system is presented as a multi-stage pipeline that combines data-driven and hand-crafted forensic techniques. It seeks to comprehensively analyze digital images by gathering a number of forensic features, combining these features, and deciding if the image is original or not. The steps are described in greater detail below.

1. Input Image Acquisition: The dataset acquired for this multi-scoring model was primarily of three origins. Majority of the images were sourced from the CASIA v1. Image tampering dataset. Complementing these samples were original and edited pictures of the research team. The CASIA dataset contained both copy-move and splicing manipulations to give the model a variety of image tampering techniques.

2. Forensic Feature Extraction: A central part of the pipeline involves extracting features using six independent forensic modules, each engineered to target a specific signature left by image manipulations. These include:

- *Metadata Analysis:* This module parses the embedded EXIF metadata to detect anomalies such as missing fields or editing tool traces, which often suggest tampering.
- *Error Level Analysis (ELA):* ELA evaluates re-compression artifacts by exaggerating differences in compression levels, making manipulated regions visibly stand out.
- *Quantization Artifact Detection:* This analyzes JPEG quantization tables to uncover inconsistencies in compression, which typically arise when different parts of an image are saved under different conditions.
- *Noise Inconsistency Analysis:* This module focuses on the spatial distribution of image noise. Natural images have consistent sensor noise, while forged regions tend to disrupt this pattern.
- *Frequency Domain Analysis:* Using Discrete Cosine Transform (DCT), this technique shifts the image to frequency space to detect unnatural periodicity often introduced during splicing or copy-move forgeries.
- *Illumination Consistency:* By analyzing shadows and light directions, this module detects physical inconsistencies that could indicate the presence of synthetic or relocated image elements.

3. Feature Fusion: The outputs of the above modules are numerical values or confidence scores indicating the likelihood of tampering from different forensic perspectives. These values then form a feature vector. This process ensures that the system captures diverse evidence types and benefits from the complementary strengths of each forensic module.

4. Dimensionality Reduction using PCA: The fused feature vector is high-dimensional and may contain correlated or redundant components. Principal Component Analysis (PCA) is applied to project this feature vector onto a lower-dimensional

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prediction while preserving the variance. This step improves computational efficiency and reduces the risk of overfitting.

5. Decision Output: In the final stage, the trained SVM model outputs a prediction label for each test image: either *forged* or *authentic*. To aid interpretability and provide transparency, the system also generates a chart visualization that breaks down the individual scores from all forensic modules. This assists in understanding which forensic cues contributed most to the final decision and offers insight for further manual investigation if necessary.

IV. Methodology

A. Pipeline Overview

The tool analyzes a list of images through six forensic techniques: metadata inspection, error level analysis (ELA), JPEG quantization table parsing, wavelet noise inconsistency detection, DFT/DCT frequency analysis, and lighting inconsistency detection. Each image receives a score from each method, which is aggregated and processed via Principal Component Analysis (PCA) to generate a unified suspicion score for forgery classification.

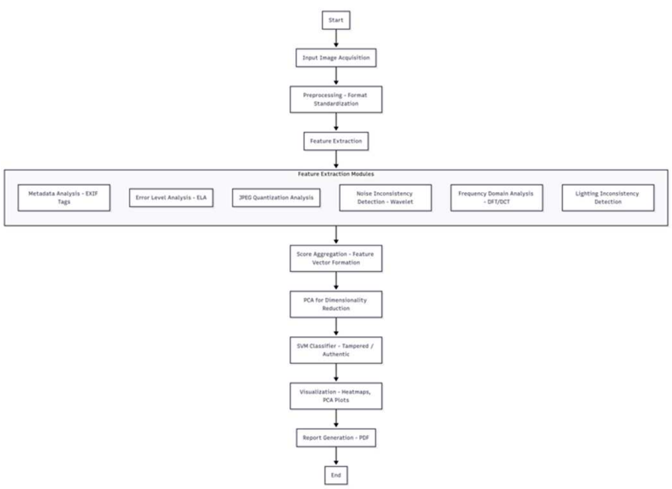

Fig. 1. Flowchart of Multi-Feature Scoring Model

B. Metadata Inspection

The tool uses MATLAB's `imfinfo()` function to inspect the EXIF metadata of each image. It checks for the presence of specific tags, such as `DateTimeOriginal`. If these tags are missing or appear unusual, the image is assigned a higher suspicion score for potential tampering.

C. Error Level Analysis (ELA)

In this step, each image is compressed and then compared with the original. The absolute differences between the two are computed, and significant differences suggest tampering. The image is resized to 80 percent of its original size, simulating compression, and the differences are computed between the original and the resized image. A higher difference score indicates potential manipulation.

D. JPEG Quantization Table Analysis

For JPEG images, the tool analyzes the quantization noise to detect signs of double or multiple compressions. If the quantization table exhibits unusual variation, it suggests that the image has been recompressed, raising suspicion of possible forgery.

E. Wavelet Noise Inconsistency

Each image is decomposed using 'Haar' wavelets. The high-frequency subbands, such as `cH` and `cV`, are analyzed to detect inconsistent noise energy. If these regions exhibit unexpected noise behavior, the image may be forged.

F. DFT/DCT Frequency Analysis

A 2-Dimensional Discrete Cosine Transform (DCT) is applied to the grayscale version of the image. The energy in the higher frequencies is analyzed, as inconsistency in the frequency domain (especially in high frequencies) could indicate manipulation.

G. Lighting Inconsistency Detection

A Sobel operator is used to calculate the image's gradients. The magnitude of the gradients is computed, and if large gradients or multiple light directions are detected, this may suggest manipulation in the image, particularly in cases of splicing or lighting inconsistencies.

```
[language=Matlab, caption=Lighting Gradient Magnitude Calculation] [Gx, Gy] = gradient(double(gray_img)); gradient_mag = sqrt(Gx.^2 + Gy.^2); max_ grad = max(gradient_mag(:));
```

H. Score Fusion via PCA

Individual scores for each forensic method are aggregated into a matrix. Principal Component Analysis (PCA) is then applied to extract the first principal component, which is treated as a unified suspicion score for each image. The PCA scores are then normalized to range from 0 to 1, where higher values indicate more suspicion of forgery. PCA helps to determine which of the six forensic methods is most useful or appropriate in its analysis, thus reducing the number of feature vectors used to determine the score.

I. Forgery Classification

Images with normalized PCA scores above 0.33 are classified into forgery types such as *Splicing*, *Recompression*, *Lighting Mismatch*, or *Copy-Move*, based on the scoring patterns observed in individual forensic methods. This classification is done using a decision rule that combines the scores from each method.

J. Report Generation and Visualization

For each image, composite heatmaps and overlays are created to visually represent the forensic results. A full PDF report is generated using MATLAB's Report Generator, which includes the original images, heatmaps, PCA scores, and final classifications.

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V. Evaluation and Results

A. ELA Heatmap Example

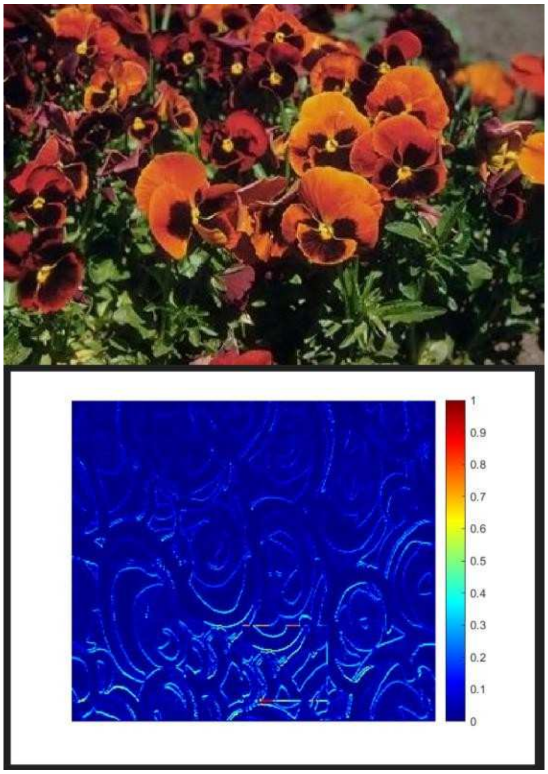

Fig. 2. Error level analysis (ELA) visualization showing original image (top) and tampering artifacts in compressed regions (bottom)

Error Level Analysis Heatmap Example

Error Level Analysis (ELA) is demonstrated in Figure 2. The original image is shown in the top image, and areas of possible tampering are highlighted in the bottom image. Areas with abnormal recompression levels, which are frequently suggestive of digital manipulation, are represented by the brighter regions in the ELA heatmap. Unreliable compression artifacts created during splicing, object insertion, or selective editing can be found with the aid of this visualization. The ELA step is especially helpful for identifying minute changes that are invisible to the naked eye.

B. PCA Score Graph

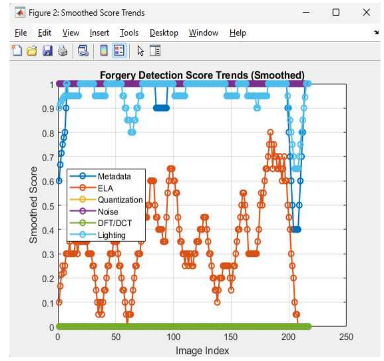

Fig. 3. PCA plot illustrating the separation between real and fake images in reduced dimensional space.

Graph of PCA Score

The pattern of distribution of PCA scores for processed images is shown in Figure 3. The multi-feature forensic scores of the images are used to map them into a reduced feature space. The difference between tampered and authentic images is evident, with tampered images typically obtaining higher PCA scores. The classifier's decision boundary is strengthened by this dimensionality reduction, which also helps reveal patterns hidden by individual feature scores. The ability of PCA to combine various forensic indicators into a single suspicion score is validated by the obvious clustering.

VI. Discussion

The results validate the effectiveness of combining multiple forensic techniques. PCA enhances interpretability of the feature vectors and improves classification performance. Additionally, x and y features proved to be the most reliable indicators of image integrity. The modular nature of the framework ensures adaptability for integration with more advanced techniques in the future.

A. Ethical Considerations

The use of image forensics must respect individual privacy and legal boundaries in each and every geographic context. While detecting tampered images is vital in journalism, law enforcement, cybersecurity and other fields relevant to social media or public perception, it is equally important to ensure that no unauthorized surveillance or data capture from unauthorized sources occurs during the process.

B. Threats to Validity

- Tool Limitations: Some expertly forged images (possi- bly using AI image generation) may bypass detection, especially if multiple forgeries cancel out each other's traces.
- Threshold Sensitivity: Risk level categorization is based on empirical thresholding, which might not be optimal for all use cases.

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VII. Future Work

To build upon this research, we propose the following directions for future development:

- Integration of DL techniques, such as Convolutional Neural Networks (CNNs), for automatic feature extraction.
- Developing a real-time browser plugin that flags suspicious images upon loading.
- Implementation of a mobile application for on-device image integrity checks.
- Optimization of our current MATLAB implementation for large-scale and real-time processing.

VIII. Conclusion

This paper proposed a comprehensive MATLAB-based pipeline to detect fake images using six forensic techniques. The combination of handcrafted forensic features, PCA, and SVM proved highly effective in real world settings. The proposed solution is scalable, modular, and can serve as a base for more advanced AI-integrated forgery detection frameworks.

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